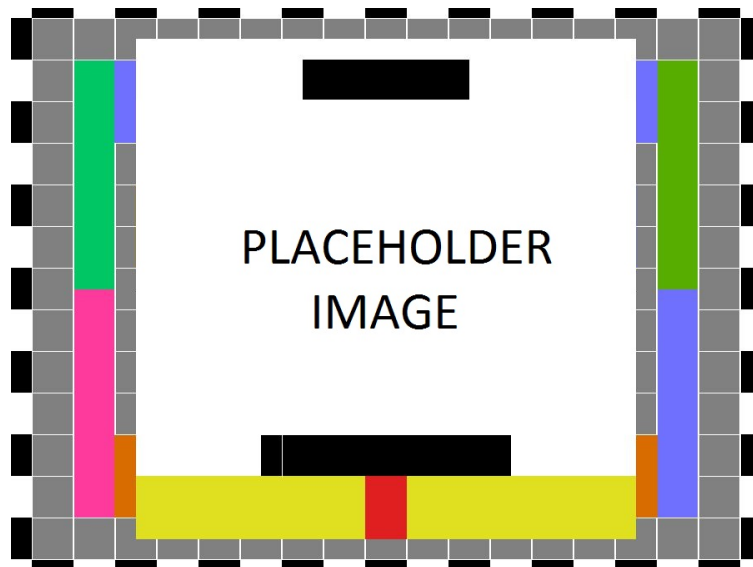


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM NIMBUS
AQUAGUARD SMART WATER-LEAK MONITOR**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	10.01.2026	JVN	document creation
1.0	10.20.2026	AT, GH, CB	official release

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1 INTRODUCTION

This SRS specifies AquaGuard Smart Water-Leak Monitor, a BLE leak sensor with a companion mobile app. It follows ISO/IEC/IEEE 29148:2018 [1].

1.1 PURPOSE

Agreed requirements baseline for Team Nimbus and its advisor.

1.2 SCOPE

The system shall detect moisture/abnormal flow at the device, alert the paired phone locally over BLE, and deliver remote push notifications via a cloud backend when the phone is away; the app shall pair, configure thresholds, and show event history. Whole-home water shut-off actuation is out of scope.

1.3 PRODUCT OVERVIEW

1.3.1 PRODUCT PERSPECTIVE

Three cooperating elements: the sensor device (BLE peripheral), the mobile app (BLE central), and a lightweight cloud backend (accounts + push).

1.3.2 PRODUCT FUNCTIONS

Leak/flow sensing; BLE pairing/config; local + remote alerting; event history; battery/status reporting; firmware update (OTA, stretch).

1.3.3 USER CHARACTERISTICS

A general homeowner using a personal smartphone; no technical expertise assumed.

1.3.4 ASSUMPTIONS & DEPENDENCIES

Recent iOS/Android phones; BLE range ≤ 5 m; free-tier push service; controlled water test rig.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Term	Definition
BLE	Bluetooth Low Energy
Peripheral/Central	BLE device roles (device advertises; phone connects)
GATT	Generic Attribute Profile (BLE service/characteristic model)
Leak event	A moisture/flow reading crossing the configured threshold

2 REQUIREMENTS ENGINEERING APPROACH

Identifiers [TYPE] - [NN] (EI, FR, US, PE, SA, SE, MS, QA) with the 29148 attribute set. Requirements are singular and verifiable.

3 EXTERNAL INTERFACE REQUIREMENTS

3.1 EI-01: BLE GATT PROFILE

DESCRIPTION

The device shall expose a BLE GATT service with characteristics for live reading, leak-event notification, threshold configuration, and battery level.

RATIONALE

The app and device must share a well-defined wireless contract.

SOURCE

Team.

PRIORITY

Critical

VERIFICATION METHOD

Test — read/write/subscribe each characteristic from the app.

CONSTRAINTS

Bluetooth SIG GATT constraints; MTU limits.

APPLICABLE STANDARDS

Bluetooth Core Specification (BLE).

3.2 EI-02: PUSH NOTIFICATION SERVICE

DESCRIPTION

The backend shall deliver leak alerts to the app via the platform push services (APNs/FCM).

RATIONALE

Alerts must reach the user when the app is backgrounded or the phone is away.

SOURCE

Team.

PRIORITY

High

VERIFICATION METHOD

Test — trigger a leak; receive a push with app backgrounded.

CONSTRAINTS

Depends on APNs/FCM availability.

APPLICABLE STANDARDS

APNs/FCM specifications.

4 FUNCTIONAL REQUIREMENTS

4.1 FR-01: LEAK DETECTION

DESCRIPTION

The device shall raise a leak event within 5 s of moisture or abnormal flow crossing the configured threshold.

RATIONALE

Early detection is the product's core purpose.

SOURCE

Advisor; team.

PRIORITY

Critical

VERIFICATION METHOD

Test — apply water on the rig; measure detection time.

CONSTRAINTS

Threshold configurable via EI-01.

APPLICABLE STANDARDS

None.

4.2 FR-02: BLE PAIRING & CONFIGURATION

DESCRIPTION

A user shall pair the device with the app and set the leak threshold through the app UI.

RATIONALE

The device must be set up without special tools.

SOURCE

Advisor.

PRIORITY

Critical

VERIFICATION METHOD

Demonstration — pair and set a threshold.

CONSTRAINTS

One device per app instance in the prototype.

APPLICABLE STANDARDS

None.

4.3 FR-03: LOCAL ALERT

DESCRIPTION

On a leak event while connected, the app shall alert the user within 3 s with a distinct visual and audible/haptic notification.

RATIONALE

Immediate on-phone warning when the user is nearby.

SOURCE

Advisor.

PRIORITY

Critical

VERIFICATION METHOD

Test — trigger event; measure alert latency.

CONSTRAINTS

Requires an active BLE connection.

APPLICABLE STANDARDS

None.

4.4 FR-04: REMOTE ALERT

DESCRIPTION

When the phone is not connected to the device, a leak event shall be delivered as a remote push notification within 30 s of reconnection/relay.

RATIONALE

Users are often away from the sensor; alerts must still reach them.

SOURCE

Advisor.

PRIORITY

High

VERIFICATION METHOD

Test — background the app; trigger event; receive push.

CONSTRAINTS

Depends on connectivity and EI-02.

APPLICABLE STANDARDS

None.

4.5 FR-05: EVENT HISTORY

DESCRIPTION

The app shall display a timestamped history of leak events and battery status for the paired device.

RATIONALE

Users need to review and trust past behavior.

SOURCE

Advisor.

PRIORITY

Moderate

VERIFICATION METHOD

Demonstration — view history after several events.

CONSTRAINTS

History retained locally and in the backend.

APPLICABLE STANDARDS

None.

5 USABILITY REQUIREMENTS

5.1 US-01: GUIDED SETUP

DESCRIPTION

A first-time user shall complete pairing and threshold setup in under 5 minutes following the in-app guide in ≥ 8 of 10 trials.

RATIONALE

Consumer devices must be self-installable.

SOURCE

Advisor.

PRIORITY

High

VERIFICATION METHOD

Test — moderated setup trials.

CONSTRAINTS

On both iOS and Android.

APPLICABLE STANDARDS

WCAG 2.2 AA (mobile).

6 PERFORMANCE REQUIREMENTS

6.1 PE-01: BATTERY LIFE

DESCRIPTION

The device shall operate at least 6 months on one battery set under nominal duty (advertising + occasional connection).

RATIONALE

Frequent battery changes would make the product impractical.

SOURCE

Advisor.

PRIORITY

High

VERIFICATION METHOD

Analysis — power budget + accelerated measurement.

CONSTRAINTS

Coin/AA battery capacity.

APPLICABLE STANDARDS

None.

6.2 PE-02: DETECTION RELIABILITY

DESCRIPTION

On the controlled test rig, leak detection shall achieve $\geq 95\%$ true-positive rate with ≤ 1 false alarm per 24 h idle.

RATIONALE

Missed leaks defeat the purpose; nuisance alarms erode trust.

SOURCE

Advisor; team.

PRIORITY

High

VERIFICATION METHOD

Test — repeated leak/idle trials on the rig.

CONSTRAINTS

Limited to tested leak scenarios.

APPLICABLE STANDARDS

None.

7 SAFETY REQUIREMENTS

7.1 SA-01: ELECTRICAL & WATER SAFETY

DESCRIPTION

The device shall isolate its electronics from the moisture pad and operate at safe low voltage (battery); no mains connection.

RATIONALE

Placing electronics near water demands isolation to avoid hazard.

SOURCE

Team; safety review.

PRIORITY

Critical

VERIFICATION METHOD

Inspection — enclosure/isolation review.

CONSTRAINTS

Battery-only design (charter constraint).

APPLICABLE STANDARDS

IEC 60950/62368 (guidance).

8 SECURITY & PRIVACY REQUIREMENTS

8.1 SE-01: SECURE PAIRING & TRANSPORT

DESCRIPTION

BLE pairing shall use a bonded, encrypted link, and app↔backend traffic shall use TLS 1.3.

RATIONALE

Prevents spoofed devices and interception of alerts/account data.

SOURCE

Team; security review.

PRIORITY

Critical

VERIFICATION METHOD

Test — verify BLE bonding; TLS scan.

CONSTRAINTS

BLE security modes supported by the module.

APPLICABLE STANDARDS

Bluetooth LE Security; RFC 8446.

9 MAINTENANCE & SUPPORT REQUIREMENTS

9.1 MS-01: DIAGNOSTICS

DESCRIPTION

The app shall surface device firmware version, battery level, and last-seen time to aid support.

RATIONALE

Enables remote troubleshooting.

SOURCE

Team.

PRIORITY

Moderate

VERIFICATION METHOD

Demonstration.

CONSTRAINTS

Fields exposed via EI-01.

APPLICABLE STANDARDS

None.

10 OTHER QUALITY ATTRIBUTES

10.1 QA-01: CROSS-PLATFORM PARITY

DESCRIPTION

Core functions (pair, configure, local alert, history) shall behave equivalently on iOS and Android.

RATIONALE

Users on either platform must get the same protection.

SOURCE

Advisor.

PRIORITY

High

VERIFICATION METHOD

Test — run the core test suite on both platforms.

CONSTRAINTS

Platform BLE-background differences documented.

APPLICABLE STANDARDS

ISO/IEC 25010:2023 (portability) [2].

11 VERIFICATION & TRACEABILITY

Req ID	Requirement (abbrev.)	Method	Acceptance
FR-01	Leak detection ≤ 5 s	T	measured ≤ 5 s
FR-03	Local alert ≤ 3 s	T	measured ≤ 3 s
FR-04	Remote push	T	push received backgrounded
PE-01	6-month battery	A	power budget meets target
SE-01	Secure pairing/TLS	T	bonded BLE; TLS1.3

12 FUTURE ITEMS

12.1 FI-01: AUTOMATIC WATER SHUT-OFF

DESCRIPTION

An add-on valve actuator that shuts off water on a confirmed leak.

RATIONALE

High value but exceeds prototype scope and budget.

SOURCE

Advisor.

PRIORITY

Future

VERIFICATION METHOD

Test (if implemented).

CONSTRAINTS

Requires plumbing actuator + safety work.

APPLICABLE STANDARDS

None.

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.
- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.